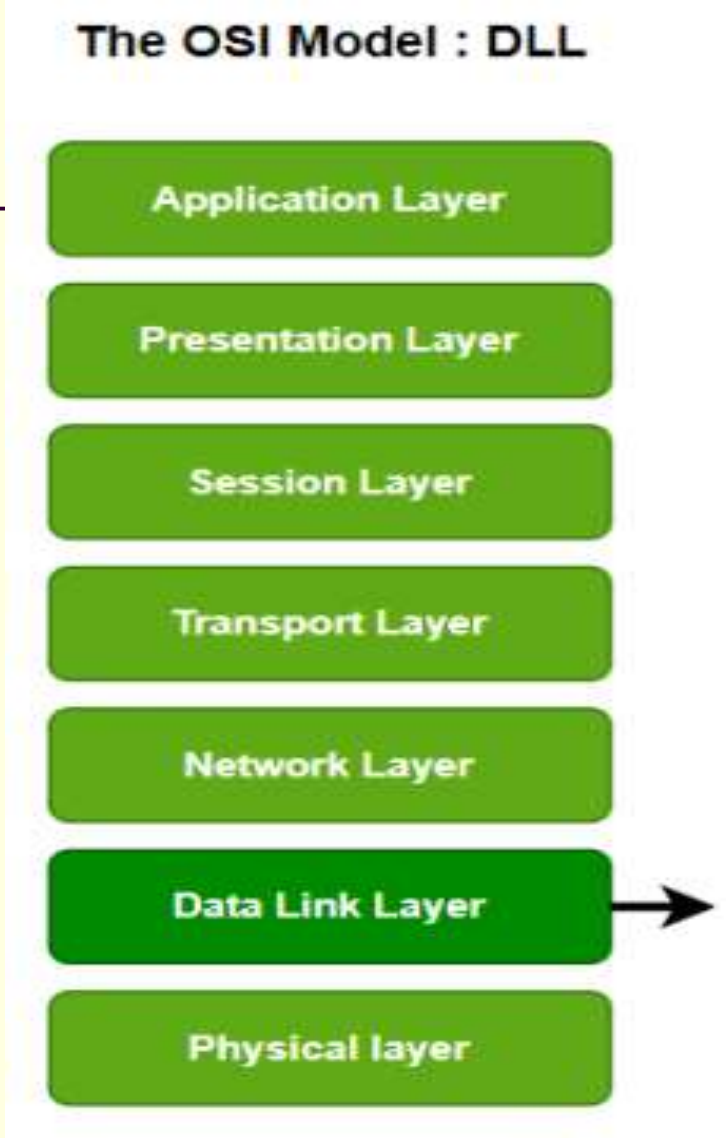
Data Link Layer

- The data link layer is the second layer from the bottom in the OSI (Open System Interconnection) network architecture model.
- It is responsible for the node-to-node delivery of data. Its major role is to ensure error-free transmission of information.
- DLL is also responsible for encoding, decoding, and organizing the outgoing and incoming data.
- This is considered the most complex layer of the OSI model as it hides all the underlying complexities of the hardware from the other above layers.



Sub-layers of Data Link Layer

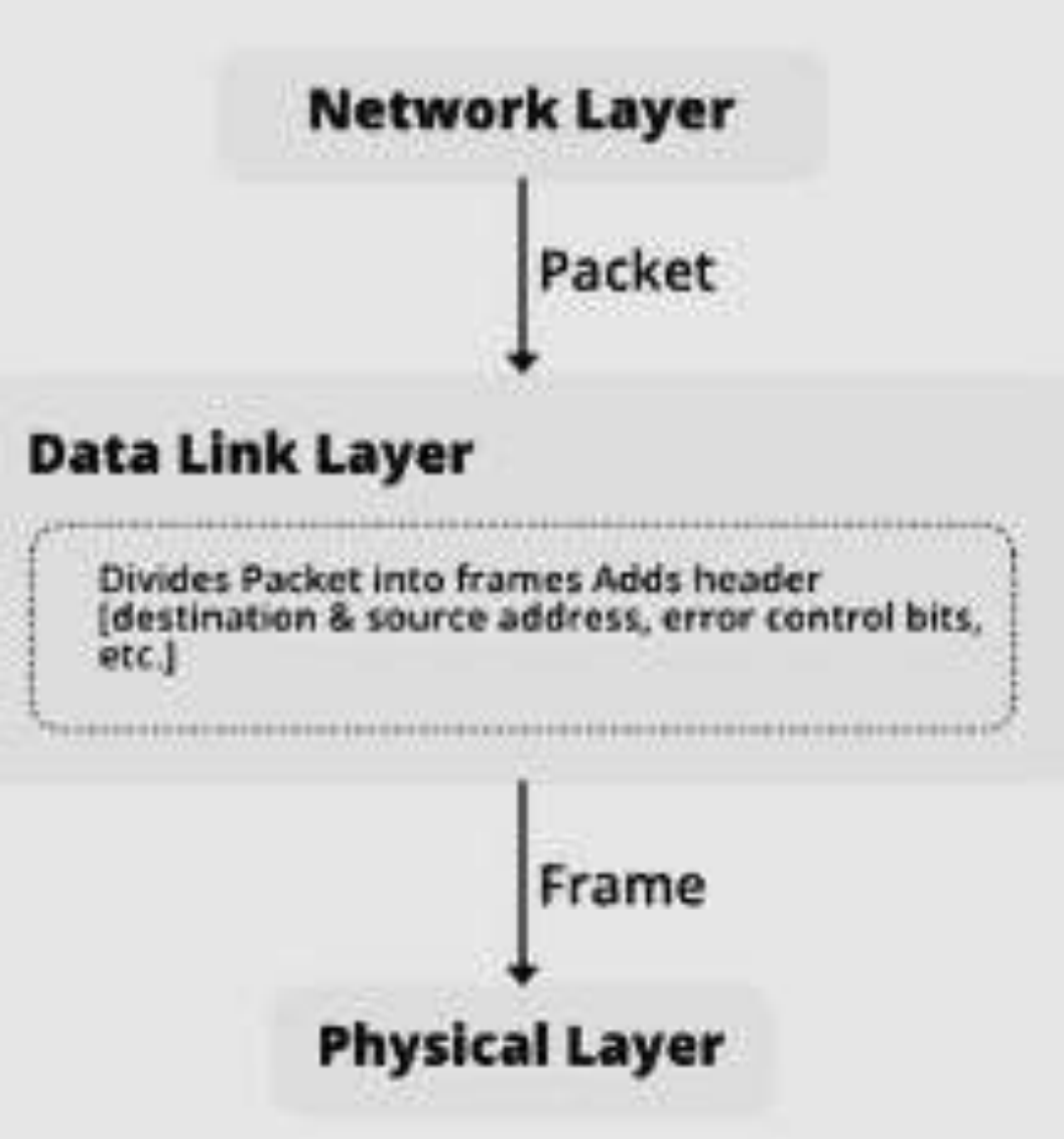
- Sub-Layers of The Data Link Layer
- The data link layer is further divided into two sub-layers, which are as follows:

- Logical Link Control (LLC)
- This sublayer of the data link layer deals with multiplexing and flow of data among applications. It is also responsible for providing error messages and acknowledgments as well.

- Media Access Control (MAC)
- MAC sublayer manages the device's interaction, responsible for addressing frames, and also controls physical media access.

- The data link layer receives the information in the form of packets from the Network layer, it divides packets into frames and sends those frames bit-by-bit to the underlying physical layer.

Functions of The Data-link Layer



Functions of The Data-link Layer

- Functions of The Data-link Layer: There are various functions of data link layers, such as:
- Framing
- The packet received from the Network layer is converted into a frame in the Data link layer. At the sender's side, DLL receives packets from the Network layer and divides them into small frames, then, sends each frame bit-by-bit to the physical layer. It also attaches some special bits (for error control and addressing) at the header and end of the frame. At the receiver's end, DLL takes bits from the Physical layer organizes them into the frame, and sends them to the Network layer.

- Addressing
- The data link layer encapsulates the source and destination's MAC address / physical address in the header of each frame to ensure node-to-node delivery. MAC address is the unique hardware address that is assigned to the device while manufacturing.

Functions of The Data-link Layer

- Error Control
- Data can get corrupted due to various reasons like noise, attenuation, etc. So, it is the responsibility of the data link layer, to detect the error in the transmitted data and correct it using error detection and correction techniques respectively. DLL adds error detection bits into the frame's header, so that receiver can check received data is correct or not. It adds reliability to physical layer by adding mechanisms to detect and retransmit damaged or lost frames.

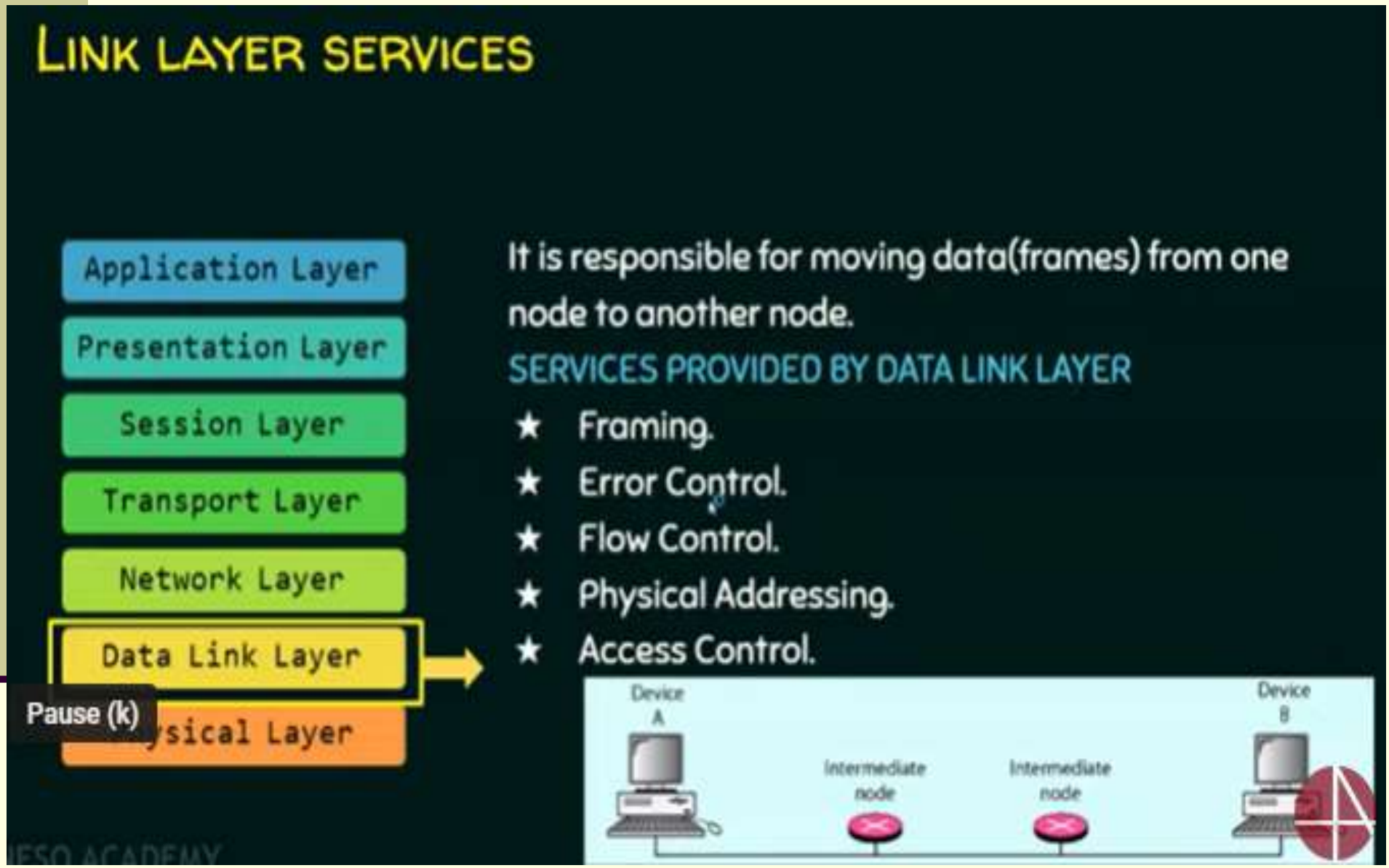
- Flow Control
- If the receiver's receiving speed is lower than the sender's sending speed, then this can lead to an overflow in the receiver's buffer and some frames may get lost. So, it's the responsibility of DLL to synchronize the sender's and receiver's speeds and establish flow control between them.

- Access Control
- When multiple devices share the same communication channel there is a high probability of collision, so it's the responsibility of DLL to check which device has control over the channel and CSMA/CD and CSMA/CA can be used to avoid collisions and loss of frames in the channel.

Protocols in the Data-link Layer

- Protocols in Data link layer
- There are various protocols in the data link layer, which are as follows:
  - Synchronous Data Link Protocol (SDLC)
  - High-Level Data Link Protocol (HDLC)
  - Serial Line Interface Protocol (SLIP)for encoding
  - Point to Point Protocol (PPP)
  - Link Access Procedure (LAP)
  - Link Control Protocol (LCP)
  - Network Control Protocol (NCP)

Flow Control

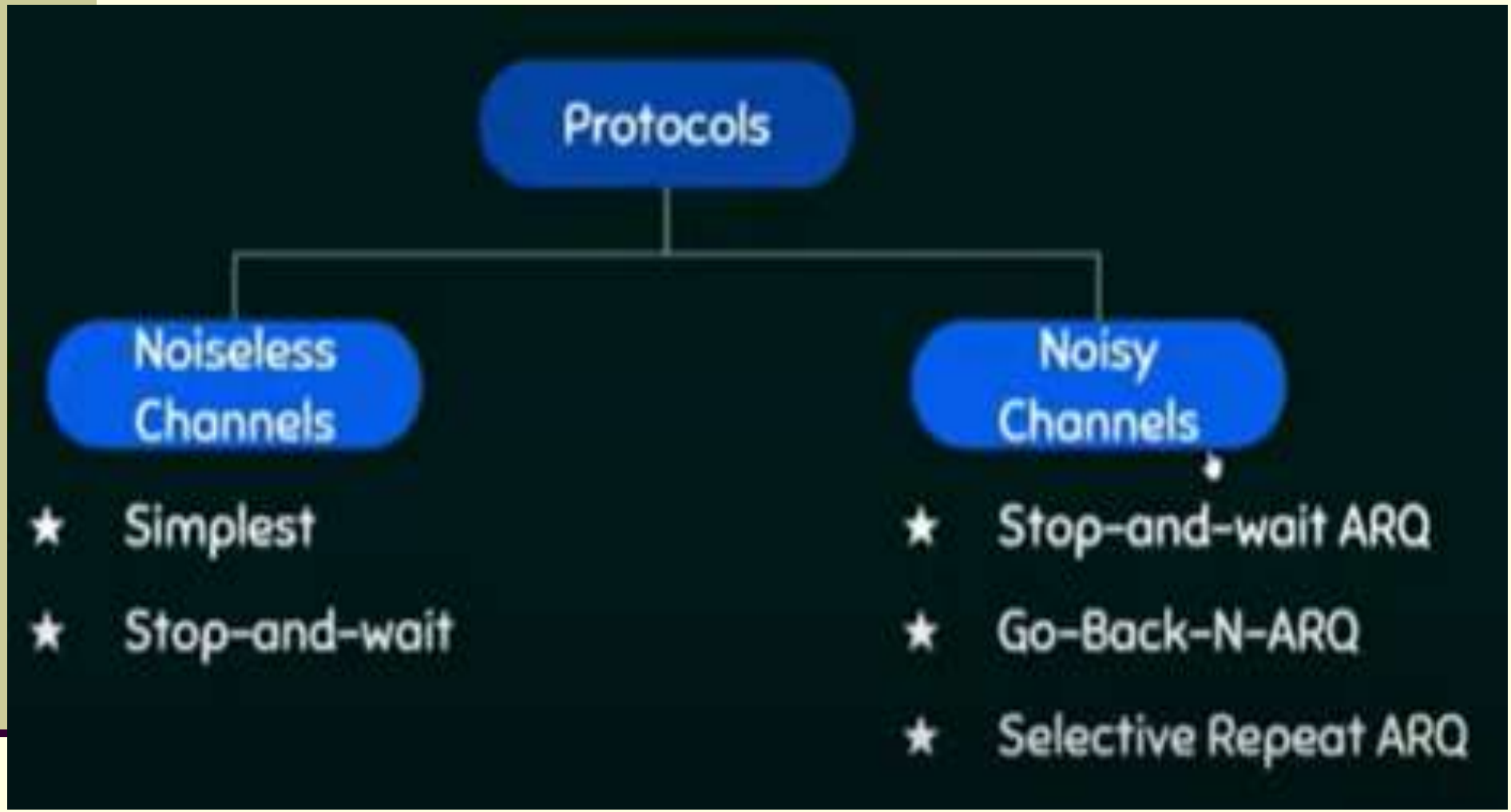


Flow Control

FLOW CONTROL

- ★ Speed matching mechanism.
- ★ Flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- ★ Flow control is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgement from the receiver.
- ★ Receiver has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- ★ Receiver must inform the sender before the limits are reached and request that the transmitter to send fewer frames or stop temporarily.

Flow Control



Flow Control

Stop-and-Wait Protocol

- ★ Stop – and – Wait protocol is data link layer protocol for transmission of frames over noiseless channels.
- ★ It provides unidirectional data transmission with flow control facilities but without error control facilities.
- ★ The idea of stop-and-wait protocol is straightforward.
- ★ After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.

Primitives of Stop-and-Wait Protocol

Sender side

Rule 1 : Send one data packet at a time.

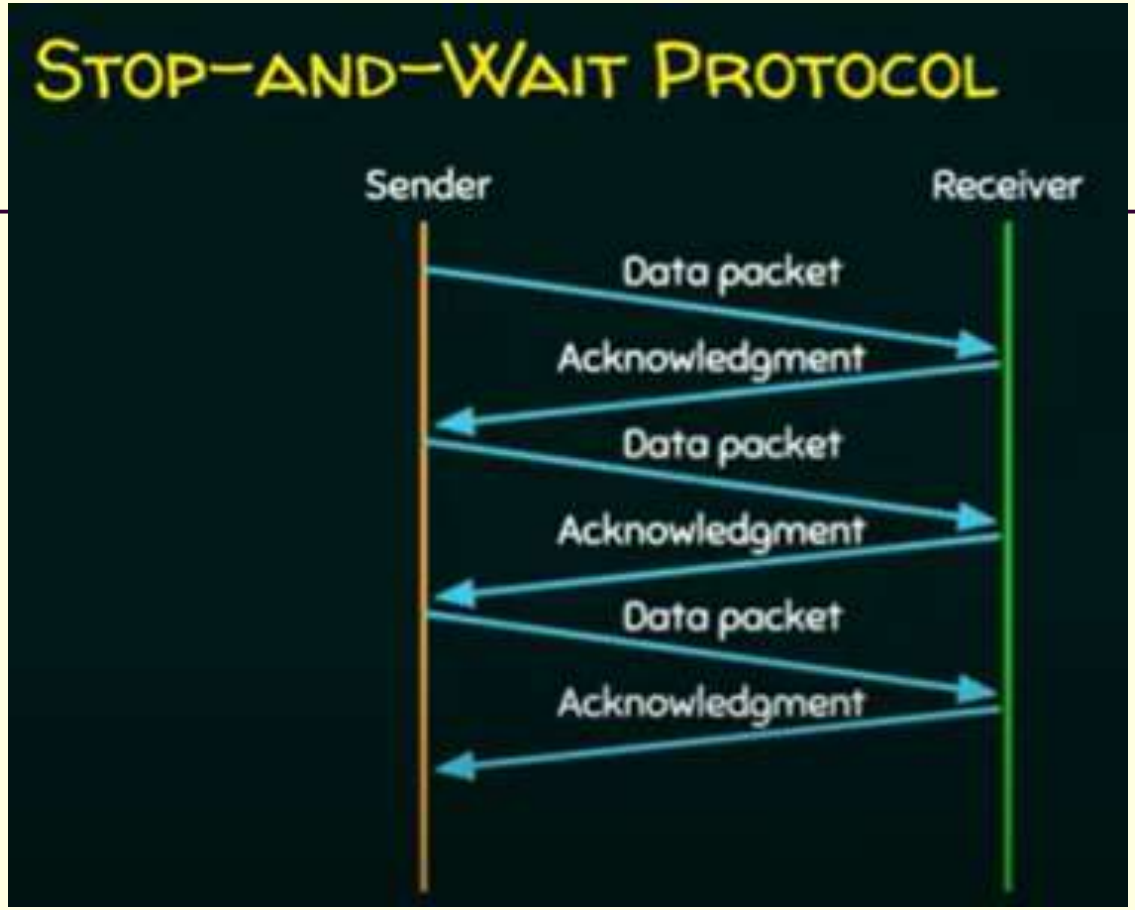
Rule 2 : Send the next packet only after receiving ACK for the previous.

Receiver side

Rule 1 : Receive and consume data packet.

Rule 2 : After consuming packet, ACK need to be sent (Flow Control).

Flow Control



- Advantage:
- Simplicity
- Disadvantage:
- For example, if there are 1000 packets, then it cannot be transmitted at same time.

Flow Control

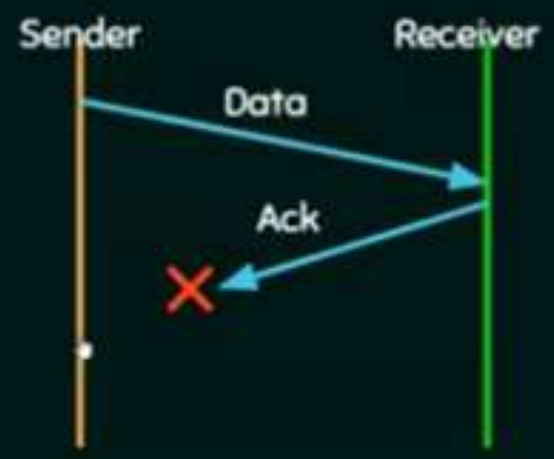
Problems of Stop-and-Wait Protocol

1. Problems due to lost data.
   - ★ Sender waits for ack for an infinite amount of time.
   - ★ Receiver waits for data an infinite amount of time.

Flow Control

Problems of Stop-and-Wait Protocol

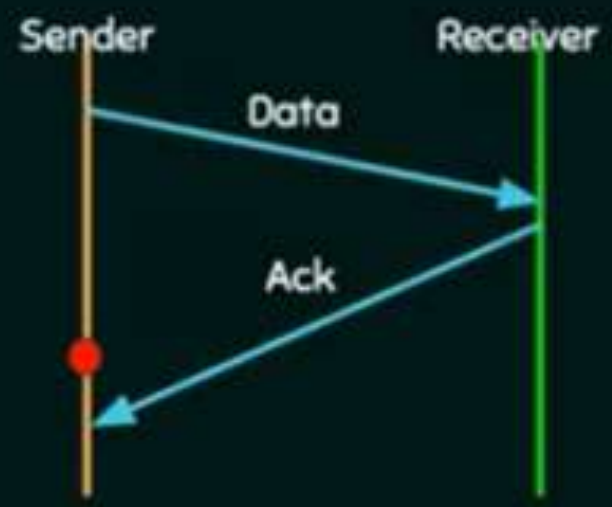
2. Problems due to lost ACK.
   - ★ Sender waits for an infinite amount of time for ack.



Flow Control

Problems of Stop-and-Wait Protocol

3. Problems due to delayed ACK/data.
   - ★ After timeout on sender side, a delayed ack might be wrongly considered as ack of some other data packet.



Flow Control

Problems of Stop-and-Wait Protocol

1. Problems due to lost data.

   Sender waits for ack for an infinite amount of time.

   Receiver waits for data an infinite amount of time.

2. Problems due to lost ACK.

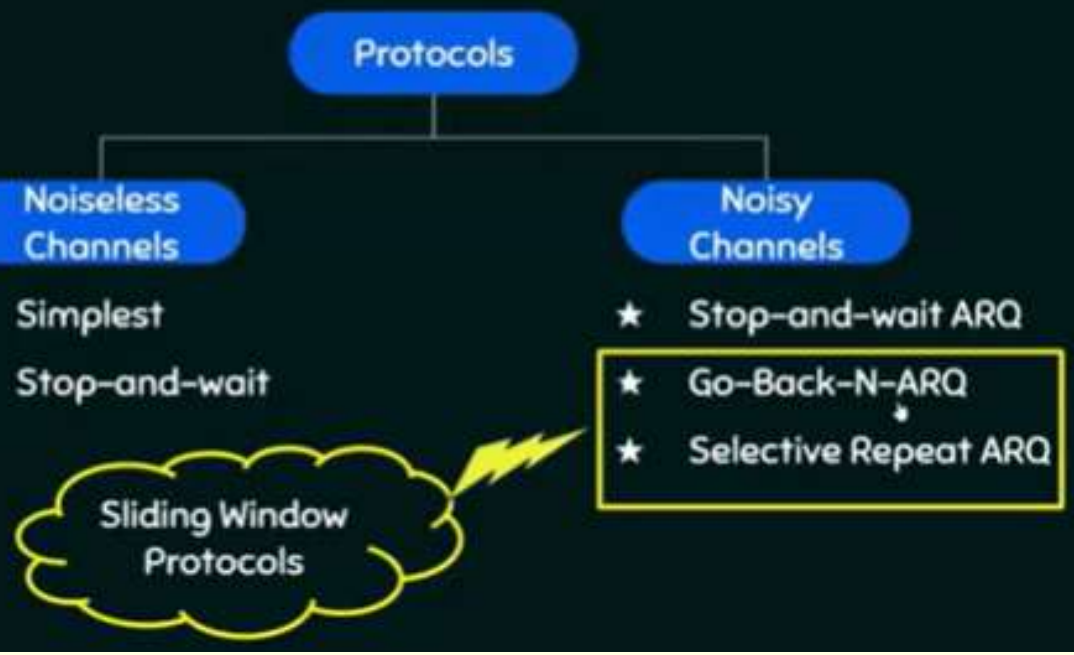
   Sender waits for an infinite amount of time for ack.

3. Problems due to delayed ACK/data.

   After timeout on sender side, a delayed ack might be wrongly considered as ack of some other data packet.

Stop and wait ARQ

Flow control



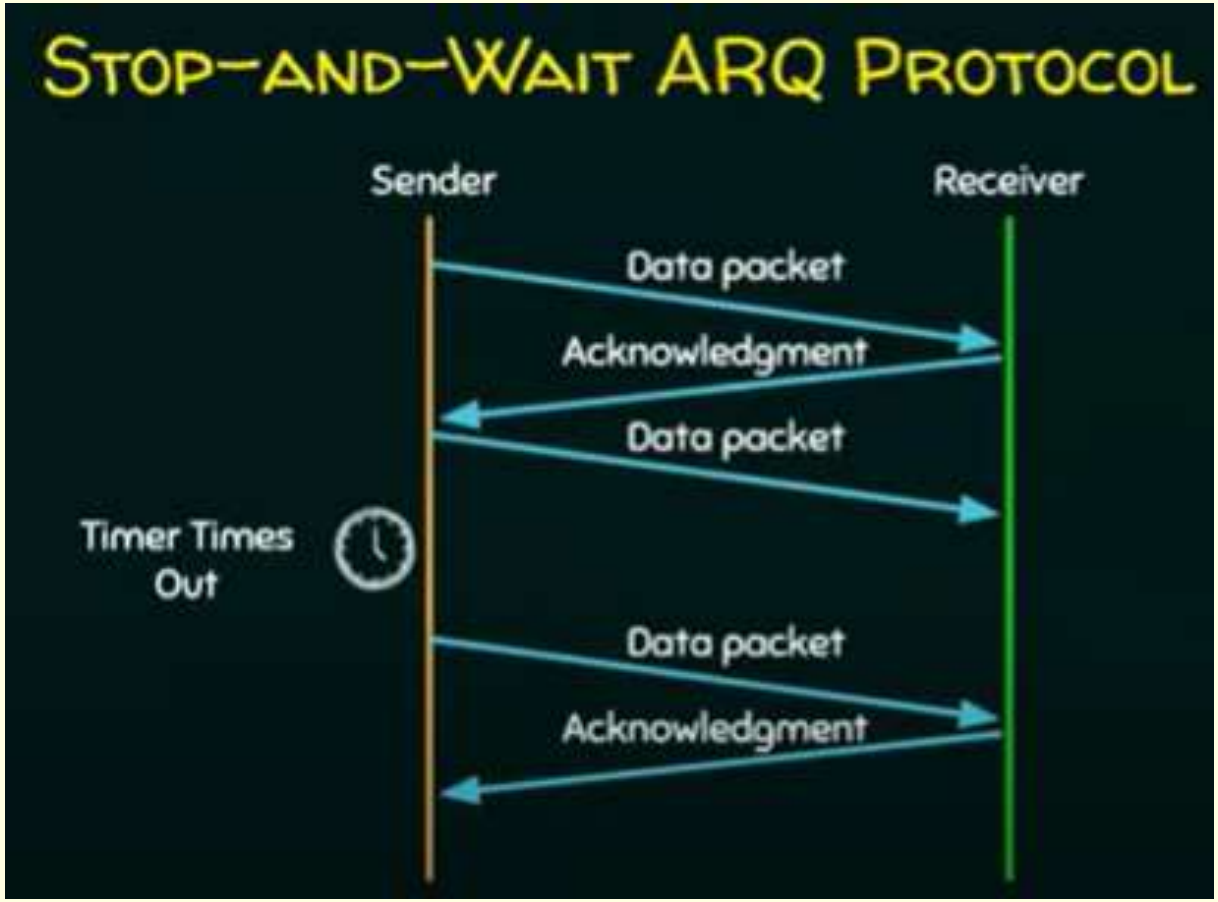
Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand the drawbacks of stop-and-wait ARQ protocol.
- ★ Know the need for sliding window protocol.
- ★ Understand the working of sliding window protocol.

Stop-and-Wait ARQ Protocol

- ★ Idea of stop-and-wait ARQ protocol is simple.
- ★ After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.
- ★ If the acknowledgement does not arrive after a certain period of time, the sender times out and retransmits the original frame.



Stop and wait ARQ

Stop-and-Wait ARQ – Drawbacks

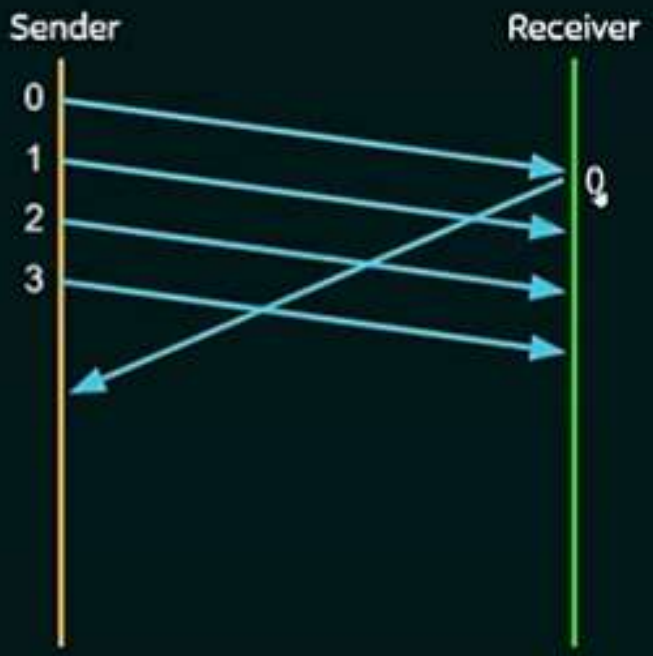
- ★ One frame at a time.
- ★ Poor utilization of bandwidth.
- ★ Poor Performance

Sliding Window Protocol

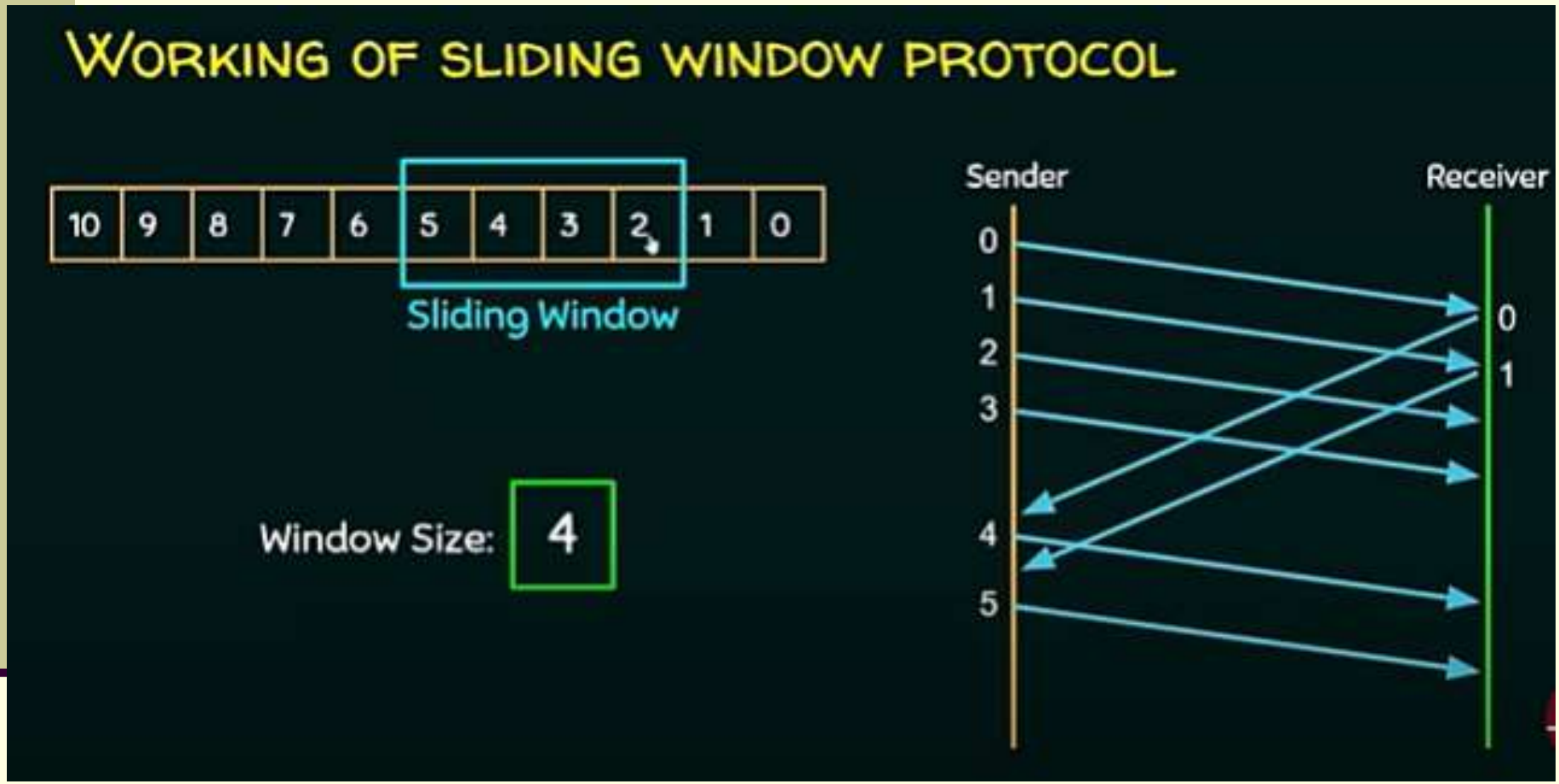
Sliding window protocol

- ★ Send multiple frames at a time.
- ★ Number of frames to be sent is based on Window size.
- ★ Each frame is numbered -> Sequence number.

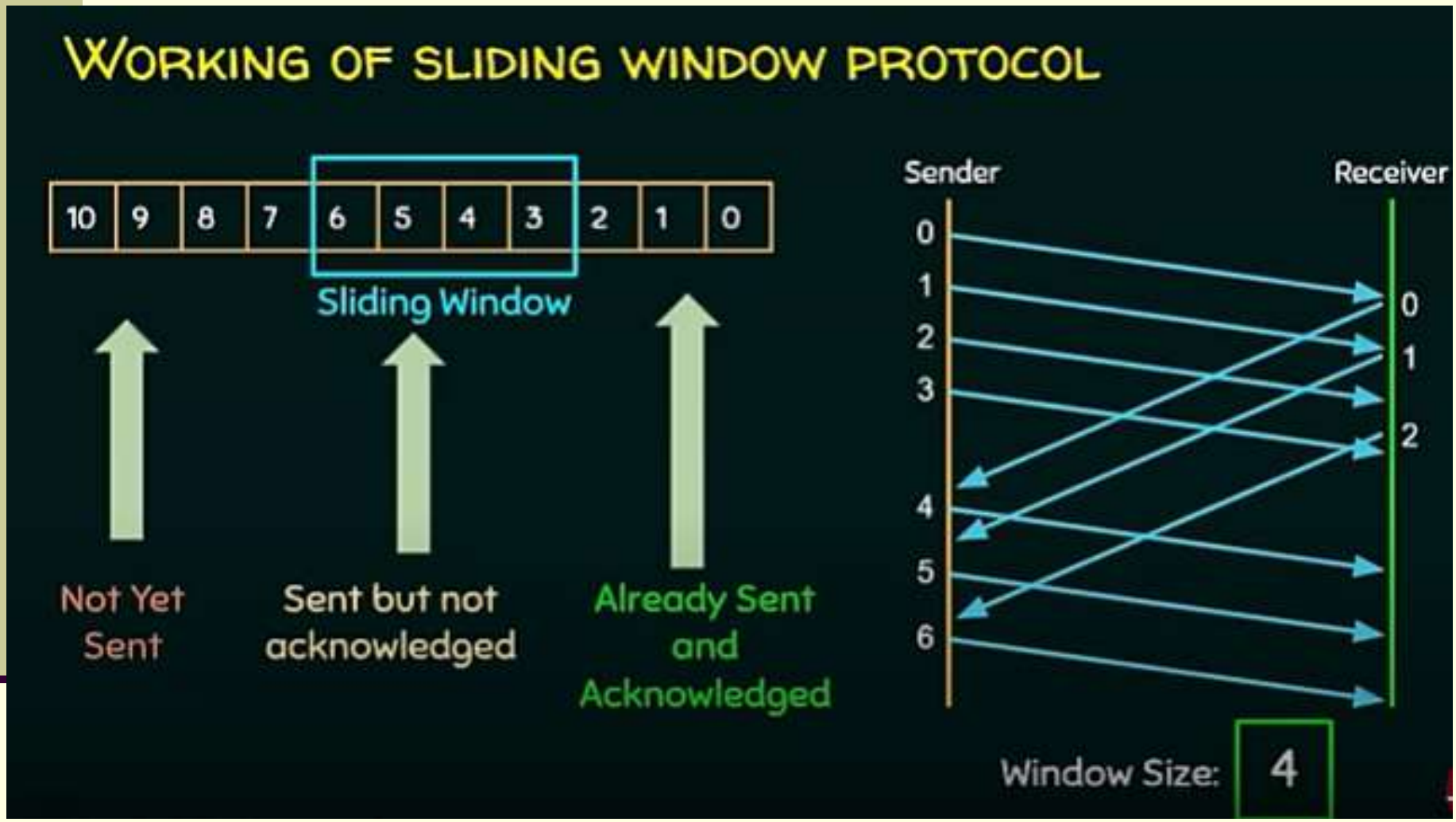
Working of sliding window protocol



Sliding Window Protocol



Sliding Window Protocol



Data Communications and Computer Networks

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